

Parameters to compare GPUs

- Introduction
 - Steps of Identify Ideal GPU for a use-case
 - Categorization of GPU
- Performance Comparison of Current Inventory
- Some Popular GPU Benchmarks
 - HPC/AI Performance Benchmarks
 - Video Rendering Performance Benchmarks
 - Analytical Processing Performance Benchmarks

Introduction [🔗](#)

To determine best GPU for a specific use case there are certain consideration that should be made like what are the dynamics for the problem, optimum/ideal performance for a solution, cost and power utilization budget, driver support for GPU, etc. The software ecosystem around GPU hardware should also be considered in making such decisions.

Steps of Identify Ideal GPU for a use-case [🔗](#)

- Identification of Primary Use Case and its requirements in terms for compute, memory and network requirements.
- Comparison of Key specification of GPUs to determine a hierarchy.
- Addition of prior use-case specific benchmarks
- Add price-to performance ratio to the GPU hierarchy.
- Check compatibility of GPU with existing code base dependencies.
- Software and ecosystem support
- Future proof

Categorization of GPU [🔗](#)

The GPUs should be clearly categorized according to thier capabilities in terms of:

- Number of parallel compute components like CUDA cores (NVIDIA) / Streaming processors (AMD).
- Memory Bandwidth, higher bandwidth can make crucial difference in inferencing servers.
- Available dedicated memory and memory type (e.g., GDDR6, HBM2).
- Availability and count of specialized cores/accelerators like **Ray Tracing** and **Tensor Cores** in NVIDIA, and **Ray Accelerators** in AMD.
- Use-case benchmarks like MLPerf/DLPerf or determination of a custom performance benchmark.
- Form factor
- Software and features: Support for features like NVIDIA's **DLSS** (Deep Learning Super Sampling) or AMD's **FidelityFX** Super Resolution, as well as driver stability and software ecosystem
- Support for APIs like DirectX, Vulkan and OpenGL

[📘 AMD Radeon RX 6800 XT Review | TechSpot](#) This kind of a comparison when drawn for HPC/AI or rendering workloads could give an intuitive incite to the user.

Performance Comparison of Current Inventory [🔗](#)

Featu res	RTX 8000	A2	A10	A30	A40	A100	L4	L40s	H100 PCIe	H100 NVL	H100 SXM	RTX A600	RTX 6000	RTX 4090
--------------	-------------	----	-----	-----	-----	------	----	------	--------------	-------------	-------------	-------------	-------------	-------------

GPUS Target Audience	0												Ada	
		Edge / Entry Level		AI Inference and Analytics		AI/HP C	Edge / Entry Level for AI and Data analysis	Generative AI/ LLMs/ Rendering						Rendering
Price(\$)	4000	1400		15000		50000	1500	6000				3500	2500	1500
Memory	48 GB GDDR6	16GB GDDR6	24GB GDDR6	24GB HBM2	48GB	40/80 GB HBM2	24GB GDDR6	48GB GDDR6 with ECC	80GB HBM2e	94GB HBM2e	80GB HBM2e	48GB GDDR6	48GB GDDR6	24GB GDDR6
Memory bus Width	384-bit	128-bit	384-bit	3072-bit	384-bit	5120-bit	192-bit	384-bit	5120-bit			384-bit	384-bit	384-bit
Memory Bandwidth (GB/s)	672	200	600	933	696	1935 2039	300	864	2000	3900	3350	768.0	768.0	1008
Memory Clock (Mz)		6251	6251	1215	7251	1512	6251	9001	1593					
GPU			GA10 2-890		GA10 2-895	GA10 0-893FF , GA10 0-893FF F, GA10 0-893H H, GA10 0-893H HH			GH10 0-200					
Clock Speed	?	1440- 1770	885- 1695	930- 1440	1305- 1740	1065- 1410	795- 2040	1065- 2520	1125- 1755			1410 MHz	2175 MHz	2235 MHz

(Base - Boost ed MHz)														
Cuda Cores	4608	1280		3rd gen		6912	7424	18,17 6				10752	18,17 6	16,38 4
Tenso r Cores	576	40 Gen 3		224		432	240	568				336	568	512
RT Cores	72	10		72		?	60	142				84	142	128
Tenso r Perfor manc e	119.4 TFLO PS	?		?		312 TFLO PS	?	1,466 TFLO PS				309.7 TFLO PS	1457. 0 TFLO PS	?
Doubl e Precis ion (FP64 FP64 Tenso r core) Pref				5.2 10.3		9.7 19.5				30 60	34 67			
FP32 Perf	14.9 TFLO PS	4.5		10.3		19.5	30.3	91.6 TFLO PS		60	67	38.7 TFLO PS	91.1 TFLO PS	82.58 TFLO PS
TF32 Perf		9 18*		82 TF 165 TF*		156 312*	120	183 366*		989	835			
FP16 (Tens or core) Perf		18 36*		165 330*		312 624*	242	362.0 5 733*		1671	1979			
FP8 Perf				-			485	733 1,466*		3341	3958			
INT8 INT4		36 144* 72 144*		330 661* 661 1321*		624 1248*	485	733 1,466* 733 1,466*		3341	3958			
RT Core	?	?		?		?	?	212 TFLO				75.6 TFLO	210.6 TFLO	?

Performance								PS				PS	PS	
Encoders	1	1		NA		NA	2	3	0	0	0	1	3	2
Decoders	1	2		4		5	4	3	7	7	7	2	3	1
Speciality	1st Ray tracing GPU	?		1 OFA 1 NVJPEG		?	4 JPEG DEC	Transformer Engine	7 JPEG DEC	7 JPEG DEC	7 JPEG DEC	?	?	?
Architecture	Turing	Amper e	Amper e	Amper e		Amper e	Ada Lovelance	Ada Lovelance	Hopper	Hoppr t	Hopper	Amper e	Ada Lovelance	Ada Lovelance
Series	Quadro	Tesla	Tesla	Tesla		Tesla	Tesla	Tesla	Tesla	Tesla	Tesla	Quadro	Quadro	GeForce
Cooling	Active	Passive	Passive	Passive		Passive	?	Passive	Passive	Passive	Passive	?	?	?
Power Consumption	250W [260-295 W]	40-60W	150W	165W		250W(40GB) . 150W-300W(80GB) . 400W(SXM)	72	350	300-350W	350-400W	700W	300W	300 W	450W
Ray Tracing	DESENT	LOW		LOW		GOOD	MEDIUM	GOOD				MEDIUM	GOOD	GOOD
AI Capabilities	LOW	LOW		MEDIUM		GOOD	MEDIUM	GOOD				MEDIUM	MEDIUM	MEDIUM
ML training	LOW	LOW		MEDIUM		GOOD	MEDIUM	GOOD				MEDIUM	MEDIUM	MEDIUM
Rendering Capabilities	LOW	LOW		LOW		GOOD	MEDIUM	GOOD				MEDIUM	MEDIUM	GOOD

Multi Instance Support	NA	NA	NA	4 MIGs @ 6GB 2 MIGs @ 12GB 1 MIGs @ 24GB		Up to 7 GPU instances	NA	NA				NA	NA	NA
NVLink / NVSwitch	Connects 2 at 100 GB/s bi-directional	NA	NA	1x 3rd Gen NVLink 200GB/s	1x 3rd Gen NVLink 112.5 GB/s	3x 3rd Gen NVLink 600GB/s	NA	NA				Connects 2 at 112.5 GB/s (bidirectional)	NA	NA
Four part ID (VID: DEVID:SSID)	10DE: 1E78: 10DE: 13D8	10DE: 25B6: 10DE: 157E	10DE: 2236: 10DE: 1482	10DE: 20B7: 10DE: 1532	10DE: 2235: 10DE: 145A	80GB - 20B5: 10DE: 1533 40GB - 10DE: 20F1: 10DE: 145F	10DE: 27B8: 10DE: 16CA	10DE: 26B9: 10DE: 1851				10DE: 2230: 10DE: 1459		
Form Factor	4.4" H x 10.5" L, FHFL, DW	HHHL, SW, (LP) PCIe	FHFL, SW	FHFL, DW	FHFL, DW	4/8 SXM GPUs in NVIDIA HGX™ A100 PCIe	1-slot low-profile, PCIe (169mm x 69mm)	FHFL, DW				FHFL, DW	FHFL, DW	?
Interface	PCIe 3.0x16	PCIe 4.0 x8, x4; 3.0 x8	PCIe 4.0 x8, x16; 3.0 x 16	PCIe 4.0 x16	PCIe 4.0 x16	PCIe 4.0 x16	PCIe 4.0 x16, x8; 3.0 x16	PCIe 4.0 x16; 64GB/s Bi	PCIe 5.0 x16, x8; 4.0 x16			PCIe 4.0 x16		

Power Connector	8-pin auxiliary power connector	-	8-pin auxiliary power connector	8-pin auxiliary power connector	8-pin auxiliary power connector	8-pin auxiliary power connector	-	16-pin auxiliary power connector	16-pin auxiliary power connector			8-pin auxiliary power connector		
------------------------	---------------------------------	---	---------------------------------	---------------------------------	---------------------------------	---------------------------------	---	----------------------------------	----------------------------------	--	--	---------------------------------	--	--

https://cloudspacetechnologies-my.sharepoint.com/:x:/g/personal/siddharth_mishra_myrealddata_in/Eff_b2M2R1pJuy42n1WO2uMB-0rqyo2iBQXOdv55_CxWEg?e=Nzoi

Connect your OneDrive account to collaborate on work across Atlassian products. [Learn more about Smart Links.](#)

OneDrive

Connect to OneDrive

 NVIDIA Data Center Platform | Line Card

	Hopper	Ampere	Turing	Volta
Supported Tensor Core precisions	FP64, TF32, bfloat16, FP16, FP8, INT8	FP64, TF32, bfloat16, FP16, INT8, INT4, INT1	FP16, INT8, INT4, INT1	FP16
Supported CUDA Core precisions	FP64, FP32, FP16, bfloat16, INT8	FP64, FP32, FP16, bfloat16, INT8	FP64, FP32, FP16, INT8	FP64, FP32, FP16, INT8

Precision format support in NVIDIA GPU Architectures

Some Popular GPU Benchmarks

HPC/AI Performance Benchmarks

- MLPerf:** A broad and widely recognized benchmark suite for machine learning performance. MLPerf covers a range of AI tasks including training and inference for different types of neural networks across various hardware platforms.
- HPL (High Performance Linpack):** Traditionally used to rank supercomputers in the TOP500 list, HPL measures a system's floating-point computing power by solving a dense system of linear equations, which is relevant for both HPC and certain AI workloads.
- HPCG (High Performance Conjugate Gradients):** Complements HPL by testing computational and data access patterns that are more characteristic of real-world HPC applications than HPL's dense linear algebra focus.
- GuideLLM:** It is a powerful tool for evaluating and optimizing the deployment of large language models (LLMs). By simulating real-world inference workloads, GuideLLM helps users gauge the performance, resource needs, and cost implications of deploying LLMs on various hardware configurations. **(For vLLM inference scenarios)**

Video Rendering Performance Benchmarks

- OctaneRender:** Uses GPU rendering to measure how well a GPU can handle photorealistic rendering using the OctaneRender engine. This is particularly relevant for professionals in visual effects and animation.
- Blender Benchmark:** Open-source 3D rendering software that offers a benchmarking tool for measuring the performance of GPUs (and CPUs) in rendering tasks. It's widely used due to Blender's popularity in the 3D modeling and animation industry.
- Redshift Benchmark:** Designed for the Redshift rendering engine, this benchmark measures the performance of GPUs in rendering scenes that are representative of motion pictures and visual effects workloads.
- V-Ray Benchmark:** V-Ray Benchmark is a free tool that measures how fast your system renders. Rendering performance evaluation can be done using CPUs, NVIDIA GPUs, or a combination of both. Chaos® V-Ray® is a 3D rendering plugin available for all major 3D design and CAD programs. It works seamlessly with 3ds Max, Cinema 4D, Houdini, Maya, Nuke, Revit, Rhino, SketchUp, and Unreal.

Analytical Processing Performance Benchmarks [🔗](#)

1. **SPECviewperf:** Widely used to evaluate the performance of GPUs in professional visualization applications, including energy, medical, and financial analysis tasks. It measures the graphics performance of systems in professional applications.
2. **Geekbench:** Provides both compute and GPU benchmarks that measure the performance of GPUs in various computational tasks, including those relevant to analytical processing.
3. **SiSoftware Sandra:** Offers a suite of benchmarks that can test various aspects of GPU performance, including processing capability, memory bandwidth, and latency, relevant for analytical processing tasks.